

PhonePay Pulse Data Analysis

Using SQL And Power BI

Project Overview

This project analyzes PhonePay Pulse transactional data using SQL And Power BI to derive business insights across payments, insurance, loans, money transfer, and recharge & bills.

Objectives

Analyze transaction trends, monitor KPIs, compare services, identify failure patterns, and support business decisions.

Tools

SQL Server, SQL, Power BI (for visualization), Excel.

SQL Concepts Used

SELECT, WHERE, GROUP BY, HAVING, ORDER BY, JOINS, Aggregate Functions, CASE, Date Functions, CTEs/Subqueries (where applicable).

KPIs

Total Transaction Amount, Total Transactions, Success vs Failed Transactions, Monthly Trends, Service-wise Performance.

Business Insights

- Monitor monthly transaction growth.
- Compare adoption across services.
- Identify failure reasons and reduce failed transactions.
- Track loan and insurance usage.
- Support dashboard-driven decision making.

Recommendations

Improve failed transaction handling, promote underused services, monitor seasonal trends, and build executive dashboards in Power BI.

Conclusion

The project demonstrates practical SQL skills for business reporting and data analysis, producing actionable insights and showcase those insights by using Power BI .

Sample Query's and Output's

- **Total Payment amount:**
Query:

```
select sum(amount) from All_transactions  
where Payment_Status='successful'
```

result:

Results Messages	
	total_payment
1	3333026090.29287

- failed payment reason's:**

Query:

```
select Reason,count(transaction_id) as total_count from All_transactions
group by Reason
having Reason<>'Successful'
```

result:

	Reason	total_count
1	Insufficient amount	3304
2	Wrong PIN	3315
3	Wrong Info	700
4	Server error	4053
5	Bank Denied	635

- monthly wise payments**

Query:

```
select datename(MONTH,date) as month_name,sum(amount) as total_amount
from All_transactions
where Payment_Status not in ('failed','Server error','Wrong PIN','Insufficient amount')
group by datename(MONTH,date)
```

result:

Results Messages		
	month_name	total_amount
1	February	268360972.332506
2	June	274776966.205576
3	August	279568492.441492
4	April	272524940.875647
5	May	284864690.652876
6	December	273837105.078888
7	January	282633151.995951
8	September	270213683.803154
9	October	282946899.46463
10	July	290782253.703545
11	November	272866226.388323
12	March	279650707.350286